Rockchip USB 2.0 Compliance Test Guide

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Preface

Overview

This document provides the USB 2.0 compliance test methodology for Rockchip platforms. According to the specifications published by the USB-IF (USB Implementers Forum), USB 2.0 compliance testing comprises the following three parts^[1]^:

- Functional Test
- Electrical Test
- Interoperability Test

This document describes the main content and test methods for each USB 2.0 test item.

Additionally, this document briefly outlines the USB certification process.

Product Version

Chip Models	Kernel Version
All Rockchip chips	All kernel versions

Target Audience

This document (this guide) is primarily intended for the following engineers:

- Technical Support Engineers
- Software Development Engineers
- Hardware Development Engineers

Revision History

Date	Version	Author	Modification Description
2020-10-09	V1.0.0	Wu Liangfeng	Initial version
2026-06-22	V1.1.0	Wang Mingcheng	Formatting fixes

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1. USB 2.0 Functional Testing

USB 2.0 functional testing primarily verifies the completeness of product functionality. The USB-IF provides the USB20CV (USB 2.0 Command Verifier) functional testing tool to evaluate whether High-Speed, Full-Speed, and Low-Speed USB devices comply with the USB Device Framework (Chapter 9), Hub Device Class (Chapter 11), HID Class, and OTG specifications. It also includes test specifications for the Mass Storage Class and USB Video Class. All USB peripherals and hubs must pass the Device Framework test to obtain certification.

The USB20CV tool supports only Windows systems. Download links are as follows:

- USB20CV 32-bit Windows
- USB20CV 64-bit Windows

Note: The USB20CV tool is supported only on Windows 7 and later versions, and the computer must support the Enhanced Host Controller Interface (EHCI).

2. USB 2.0 Electrical Characteristics Testing

USB 2.0 Electrical Characteristics Test: This test primarily evaluates the electrical signals of the product. The USB 2.0 Electrical Test Specification defines a set of electrical test standards for USB 2.0 products^[2][3]^. All USB products must pass the USB electrical characteristics test to obtain USB-IF certification.

USB 2.0 electrical characteristics testing must cover High-Speed, Full-Speed, and Low-Speed signals. The corresponding test specifications are as follows:

- Low-Speed/Full-Speed Electrical Test Specification: USB-IF Full and Low Speed Electrical and Interoperability Compliance Test Procedure
- High-Speed Electrical Test Specification: USB 2.0 Electrical Compliance Test Specification

Testing USB 2.0 electrical characteristics requires an oscilloscope from a USB-IF approved vendor:

- Keysight
- Rohde & Schwarz
- Tektronix
- Teledyne LeCroy

In accordance with USB 2.0 electrical characteristics specifications, it is recommended to execute the test items listed in Table 1 below.

Table 1: Major USB 2.0 Electrical Characteristics Test Items

Test Item	Test Requirements
Device HS Signal Quality Test	Reference [2] EL_2 EL_4 EL_5 Data Eye and Mask Test EL_6 Device Rise Time EL_6 Device Fall Time EL_7 Device Non-Monotonic Edge Test
Device HS Packet Parameters	Reference [2] EL_21 Sync Field Length Test EL_25 EOP Length Test EL_22 Measure Inter-packet Gap Between Second and Third Packets EL_22 Measure Inter-packet Gap Between First and Second Packets
Device HS Device CHIRP Timing	Reference [2] EL_28 Measure Device CHIRP-K Latency EL_29 Measure Device CHIRP-K Duration EL_31 Hi-Speed Terminations Enable and D+ Disconnect Time
Device HS Suspend/Resume/Reset Timing	Reference [2] EL_38 EL_39 Suspend Timing Response EL_40 Resume Timing Response EL_27 Device CHIRP Response to Reset from Hi-Speed EL_28 Device CHIRP Response to Reset from Suspend
Device HS Test J/K, SE0_NAK	Reference [2] EL_8 J Test EL_8 K Test EL_9 SE0_NAK Test
Device HS Receiver Sensitivity	Reference [2] EL_18 Receiver sensitivity Test - Minimum SYNC Field EL_17 Receiver sensitivity Test EL_16 Receiver sensitivity Test @ Squelch
Host HS Signal Quality	Reference [2] EL_6 Rise Time EL_6 Fall Time EL_3 Data Eye and Mask Test EL_7 Non-Monotonic Edge Test
Host HS Packet Parameters	Reference [2] EL_21 Sync Field Length Test EL_25 EOP Length Test EL_23 Inter-packet Gap Between First 2 Packets Test EL_22 Inter-packet Gap Between 2nd and 3rd Packet Test EL_55 SOF EOP Width Test

Test Item	Test Requirements
Host HS Disconnect Detect	Reference [2] EL_37 Disconnect Detect Test At 525mV Threshold EL_36 Disconnect Detect Test At 625mV Threshold
Host HS CHIRP Timing	Reference [2] EL_33 CHIRP Timing Response EL_34 CHIRP K Width EL_34 CHIRP J Width EL_35 SOF Timing Response
Host HS Suspend/Resume Timing	Reference [2] EL_39 Suspend Timing Response EL_41 Resume Timing Response
Host HS Test J/K, SE0_NAK	Reference [2] EL_8 J Test EL_8 K Test EL_9 SE0_NAK Test
Host LS/FS Inrush Current Test	Reference [3] B.4 Inrush Current Testing
Host FS Signal Quality Test	Reference [3] B.3.3.2 Full-speed Downstream Signal Quality Test
Host LS Signal Quality Test	Reference [3] B.3.3.1 Low-speed Downstream Signal Quality Test

According to Table 1, USB 2.0 electrical characteristics testing primarily focuses on high-speed signal quality. The USB-IF provides the Windows-based HSETT (High-Speed Electrical Test Tool) utility for testing USB high-speed signals. Download links are as follows:

- EHSETT 32-bit version
- EHSETT 64-bit version
- XHSETT 32-bit version
- XHSETT 64-bit version

For tool usage instructions, refer to the following document:

HSET Documentation version 0.41 for EHCI and xHCI

Note:

When performing Device/Host HS Signal Quality testing, the USB device under test must be configured into Test Packet Mode to trigger the correct test signals. Rockchip platform USB controllers support two methods for entering this mode: test commands and the HSETT tool. Specific requirements are as follows:

- USB 2.0 Device: The USB controller can be set to Test Packet Mode using either test commands or the HSETT tool.
- USB 2.0 Host: The USB controller can only be set to Test Packet Mode using test commands.

For test commands, refer to the section "USB 2.0 Tx Test Commands and Test Tools" in the document Rockchip_Developer_Guide_USB_SQ_Test_EN.

3. USB 2.0 Interoperability Testing

USB 2.0 interoperability testing primarily evaluates the compatibility of USB devices across different software operating systems and hardware controllers. Devices from the "Gold Tree" may be utilized for interoperability testing to verify compatibility between devices and various hosts, such as confirming successful enumeration of a device on a host using an EHCI controller.

Since the USB-IF does not mandate USB 2.0 interoperability testing, this document does not provide detailed specifications for such tests. For readers seeking further information on USB 2.0 interoperability test methods, please refer to the following specification:

xHCI Interoperability Test Procedures For Peripherals, Hubs and Hosts

4. USB 2.0 Certification

USB 2.0 certification refers to the process where a USB product passes specific tests defined by the USB-IF and obtains authorization to use the USB logo. There are two methods for product testing:

- Participating in a USB-IF sponsored Compliance Workshop
- Utilizing a USB-IF Authorized Test Lab (ATL)

Independent test laboratories meeting USB-IF requirements can be found at https://www.usb.org/labs. Upon passing compliance testing, the product receives a USB-IF Testing ID (TID). This TID allows verification of the test device information on the USB-IF official website and grants the right to use the USB logo.

The general process for USB compliance test certification is as follows:

- Apply for USB-IF membership to obtain a Vendor ID (VID);
- Prepare the USB compliance test checklist;
- Submit the product to a USB-IF authorized independent test laboratory for testing;
- Obtain the qualification report and certificate;
- Use the corresponding USB logo.

Note: Rockchip is a member of USB-IF with VID 0x2207. Additionally, the default VID configured in the SDK USB is also 0x2207. However, this VID is intended solely for product development and debugging purposes and cannot be licensed to developers. If a product requires USB-IF certification, the developer must independently apply for USB-IF membership to obtain a unique Vendor ID.

5. References

1. USB 2.0 Testing Information
2. USB 2.0 Electrical Compliance Test Specification
3. USB-IF Full and Low Speed Electrical and Interoperability Compliance Test Procedure
4. HSET Documentation version 0.41 for EHCI and xHCI
5. Rockchip_Developer_Guide_USB_SQ_Test_EN